



BookStoreWeb – A Web Application

- Name : **MyBookStoreWeb**
- Ports : 5000 , 5100 ,5200

Steps to Develop MyBookStoreWeb:

A)Setup the Boot Project

1) Copy BookPriceMS as BookStoreWeb

2) Remove All except the following

- a) pom.xml
- b) application.properties
- c) JLCBookPriceConfig.java
- d) StartMyBookPriceMS.java

3) Update pom.xml as given in Page No: 100

4) Update application.priperities as follows.

```
spring.application.name=MyBookStoreWeb
server.port=5000
```

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

5) Rename the Package to com.jlcindia.bookstoreweb

6) Rename Spring Bean Configuration Class

```
*** JLCBookStoreWebConfig.java
```

7) Configure following 2 beans in Bean Configuration Class

```
@Bean
public ViewResolver viewResolver() {
    InternalResourceViewResolver resolver = new InternalResourceViewResolver();
    resolver.setPrefix("WEB-INF/myjsps/");
    resolver.setSuffix(".jsp");
    resolver.setViewClass(JstlView.class);
    return resolver;
}
```

@Override

```
public void addResourceHandlers(ResourceHandlerRegistry registry) {  
    registry.addResourceHandler("/webjars/**")  
        .addResourceLocations("classpath:/META-INF/resources/webjars/");  
}
```

8) Rename BootApplication Starting Class.

*** BookStoreWebApplication.java

B)Develop the BookStore Web Application Code:

9) Write DTO's

```
*** Book.java  
*** BookInfo.java  
*** Order.java  
*** OrderItem.java  
*** OrderInfo.java  
*** UserRating.java
```

10) Write Service Layer components

```
*** BookStoreService.java  
*** BookStoreServiceImpl.java
```

11) Write Controller Layer component

```
*** BookStoreController.java
```

12) Create the Source Folder called src/main/webapp

13) Create WEB-INF Folder under src/main/webapp

14) Create the myjsps Folder under src/main/webapp/WEB-INF

15) Place the following JSP's under myjsps

```
*** index.jsp  
*** myheader.jsp  
*** myfooter.jsp  
*** showBooksList.jsp  
*** showBookInfo.jsp  
*** showCart.jsp
```

```

*** orderSuccess.jsp
*** addRating.jsp
*** ratingSuccess.jsp
*** ratingList.jsp

```

16) Create the META-INF Folder under src/main/resources

17) Create the resources Folder under META-INF

18) Create the mycss Folder under META-INF/resources

19) Add bookstore.css under META-INF/resources/mycss

20) Update bookstore.css

21) Deploy and Run

- Before Starting BookStoreWeb, make sure that All other MiceoServices are Up and Running.

BookStoreWeb: Files required

1. index.jsp	2. myheader.jsp
3. myfooter.jsp	4. showBooksList.jsp
5. showBookInfo.jsp	6. showCart.jsp
7. orderSuccess.jsp	8. addRating.jsp
9. ratingSuccess.jsp	10. ratingsList.jsp
11. bookstore.css	12. Book.java
13. BookInfo.java	14. Order.java
15. OrderItem.java	16. OrderInfo.java
17. UserRating.java	18. BookStoreController.java
19. BookStoreService.java	20. BookStoreServiceImpl.java
21. JLCBookStoreWebConfig.java	22. BookStoreWebApplication.java
23. application.properties	24. pom.xml



1.index.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<h2> <a href="showAllBooks">Display All Books</a> </h2>
  <c:import url="myfooter.jsp"/>
</body>
</html>
```

3. myfooter.jsp

```
<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

<footer class="bg-dark mytext-center">
  <span class="mytext-large mywhite">&copy; 2020 JLCBookStore</span> <br/>
</footer>
```

2.myheader.jsp

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top">
  <h4> <a href="/" class="navbar-brand"> JLCBookStore </a> </h4>

  <button class="navbar-toggler" type="button" data-toggle="collapse" data-
target="#navbarColor01" aria-controls="navbarColor01" aria-expanded="false" aria-
label="Toggle navigation">

    <span class="navbar-toggler-icon"></span>
  </button>
  <div class="collapse navbar-collapse" id="navbarColor01" align="right">
    <ul class="navbar-nav mr-auto mt-2 mt-lg-1">
      <li class="nav-item active">
        <h4> <a href="/showRatingsForm" class="nav-link">Rate the books</a> </h4>
      </li>
```



```
<li class="nav-item active">
  <h4> <a href="/showMyRatings" class="nav-link" >Show My Ratings</a> </h4>
</li>
<li class="nav-item active">
  <h4> <a href="/showMyRatings" class="nav-link" >Show My Orders</a> </h4>
</li>
<li class="nav-item active">
  <h4> <a href="/mylogin" class="nav-link" >Login</a> </h4>
</li>
<li class="nav-item active">
  <h4> <a href="/myregister" class="nav-link">Register</a> </h4>
</li>
</ul>
</div>
</nav>
```

4. showBooksList.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp" />
</div>
<div class="container">
<form action="/showAllBooks" >
  <table class="table table-striped table-bordered table-light myfont">
<tr>
<td>
  <select name="author" class="form-control mytext" >
    <c:forEach var="authorName" items="${MyAuthorList}">
      <option value="${authorName}">
        <c:out value="${authorName}" />
      </option>
    </c:forEach>
  </select>
</td>
```



```
<td>
  <select name="category" class="form-control mytext" >
    <c:forEach var="catName" items="${MyCatList}">
      <option value="${catName}">
        <c:out value="${catName}" />
      </option>
    </c:forEach>
  </select>
</td>
<td> <input class="btn btn-primary" type="submit" value="Show Books " /> </td>
</tr>
</table>
</form>
</div>

<div class="container">
  <table class="table table-striped table-bordered table-light myfont">
    <thead class="bg-info">
      <tr>
        <th>Book ID</th>
        <th>Book Name</th>
        <th>Author</th>
        <th>Publications </th>
        <th>Category</th>
        <th colspan="2" align="center">
          <a href="showMyCart" class="btn btn-danger"> Show My Cart </a>
        </th>
      </tr>
    </thead>
    <tbody>
      <c:forEach var="mybook" items="${MyBooksList}">
        <tr>
          <td> ${mybook.bookId } </td>
          <td> ${mybook.bookName }</td>
          <td> ${mybook.author }</td>
          <td> ${mybook.publications }</td>
          <td> ${mybook.category }</td>
          <td>
            <td>
              <form:form action="addToCart">
                <input type="hidden" name="bookId" value="${mybook.bookId }"/>
                <input class="btn btn-primary" type="submit" value=" Add to My Cart " />
              </form:form>
            </td>
          </td>
        </tr>
      </c:forEach>
    </tbody>
  </table>
</div>
```



```
<td>
<form:form action="showBookInfo" method="get">
<input type="hidden" name="bookId" value="{mybook.bookId }"/>
<input class="btn btn-primary" type="submit" value=" More Info " />
</form:form>
</td>

</tr>
</c:forEach>
</tbody>
</table>
</div>
<c:import url="myfooter.jsp"/>
</body> </html>
```

5. showBookInfo.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>

<body>
  <div class="container">
    <table class="table table-striped table-bordered table-light myfont">

<tr>
<th>Book ID</th>
<td> ${MyBookInfo.bookId } </td>
</tr>

<tr>
<th>Book Name</th>
<td> ${MyBookInfo.bookName }</td>
</tr>
```



```
<tr>
<th>Author</th>
<td>${MyBookInfo.author }</td>
</tr>
<tr>
<th>Publications </th>
<td>${MyBookInfo.publications }</td>
</tr>
<tr>
<th>Category</th>
<td>${MyBookInfo.category }</td>
</tr>
<tr>
<th>Price </th>
<td>Rs. ${MyBookInfo.price }</td>
</tr>
<tr>
<th>Offer </th>
<td><span class="mytext-large btn-danger"> ${MyBookInfo.offer }% Off </span></td>
</tr>
<tr>
<th>Ratings </th>
<td>${MyBookInfo.avgRating }</td>
</tr>
</table>
</div>
<div class="container" align="center">
<h1 class="text-center">
<a href="continueShopping" class="btn btn-danger"> Continue Shopping</a>
</h1>
</div>
<br/>
<c:import url="myfooter.jsp"/>
</body> </html>
```

6. showCart.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
```




```
<link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp" />
</div>
<body>

<div class="container">
  <table class="table table-striped table-bordered table-light myfont">
    <thead class="bg-info">
<tr>
<th>Book ID</th>
<th>Book Name</th>
<th>Author</th>
<th>Publications </th>
<th>Category</th>
<th>
<a href="continueShopping" class="btn btn-danger"> Continue Shopping</a>
</th>
</tr>
</thead>
<tbody>
<tr>
<td colspan="6" align="center">
${CartEmptyMSG}
</td>
</tr>
<c:forEach var="mybook" items="${MyCartItems}">
<tr>
<td> ${mybook.bookId } </td>
<td>${mybook.bookName }</td>
<td>${mybook.author }</td>
<td>${mybook.publications }</td>
<td>${mybook.category }</td>
<td>
<form:form action="removeFromCart" method="post">
<input type="hidden" name="bookId" value="${mybook.bookId }"/>
<input class="btn btn-primary" type="submit" value=" Remove from My Cart " />
</form:form>
</td>
</tr>
</c:forEach>
</tbody>
```



```
<tfoot>
<tr>
<td colspan="6" align="center">
<a href="placeOrder" class="btn btn-danger"> Place My Order</a>
</td>
</tr>
</tfoot>
</table>
</div>
<br/><br/>
<br/>
<c:import url="myfooter.jsp"/>
</body>
</html>
```

7. orderSuccess.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html> <head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<body>
  <br/><br/>
  <br/><br/>
<div class="container">
  <h1 class="text-center"> Your Order has been placed successfully </h1>
</div>
  <br/><br/>
  <br/><br/>
  <div class="container" align="center">
    <h1 class="text-center">
      <a href="continueShopping" class="btn btn-danger"> Continue Shopping</a>
    </h1>
  </div>
<br/><br/><br/><br/> <br/>
  <c:import url="myfooter.jsp"/>
</body>
</html>
```



8. addRating.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
    <title>JLC Bookstore</title>
    <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
    <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<body>
<div class="container">
<form:form action="/addMyRating" modelAttribute="myUserRating">
    <table class="table table-striped table-bordered table-light myfont">
<tr>
<th colspan="2" align="center">
Provide You Rating
</tr>
<tr> <th>User Id</th>
<td> <form:input path="userId" readonly="true"/> </td>
</tr>
<tr> <th>Book Id</th>
<td> <form:input path="bookId"/> </td>
</tr>
<tr>
<th>Rating</th>
<td> <form:input path="rating"/> </td>
</tr>
<tr>
<th>Review </th>
<td> <form:textarea path="review"/> </td>
</tr>
<tr>
<th colspan="2" align="center">
<input class="btn btn-primary" type="submit" value=" Add Rating Info " />
</th>
</tr>
</table>
</form:form>
</div>
```



```
</></></></></>
<c:import url="myfooter.jsp"/>
</body>
</html>
```

9. ratingSuccess.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<body>
  <br/><br/> <br/><br/>
<div class="container">
  <h1 class="text-center"> Your Rating Added Successfully </h1>
<br/> <br/>
  <h1 class="text-center">100 Points added BookStore Wallet </h1>
  <h1 class="text-center">Thank You </h1>
</div>
  <br/><br/> <br/><br/>
  <br/><br/> <br/>
  <c:import url="myfooter.jsp"/>
</body>
</html>
```

10. ratingList.jsp

```
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html>
<head>
  <title>JLC Bookstore</title>
  <link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
  <link href="mycss/bookstore.css" rel="stylesheet">
</head>
<body>
```



```
<div class="card">
<c:import url="myheader.jsp"/>
</div>
<body>
<div class="container">
  <table class="table table-striped table-bordered table-light myfont">
    <thead class="bg-info">
      <tr>
        <th>Book ID</th>
        <th>User ID</th>
        <th>Rating</th>
        <th>Review</th>
      </tr>
    </thead>
    <tbody>
      <c:forEach var="myuserRating" items="${MyUserRatingList}">
        <tr>
          <td> ${myuserRating.bookId } </td>
          <td> ${myuserRating.userId }</td>
          <td> ${myuserRating.rating }</td>
          <td> ${myuserRating.review }</td>
        </tr>
      </c:forEach>
    </tbody>
  </table>
</div>
  <c:import url="myfooter.jsp"/>
</body> </html>
```

11.bookstore.css

```
.myfont{
font-weight: bold;
}

.mymargin-1{
  margin: 0px 10px;
}

.myright{
text-align: right;
}

.mymargin{
```



```
margin-bottom: 10px;  
}
```

```
.myheight{  
line-height: 2.5em;  
}
```

```
.my-li-no-display{  
list-style-type: none;  
}
```

```
.myradius{  
border-radius: 15px;  
}
```

```
.hello{  
padding-left: 25px;  
padding-right: 25px;  
}
```

```
.mytext-center{  
text-align: center;  
}
```

```
.mytext-large{  
font-size:22px;  
}
```

```
.mytext-extra-large{  
font-size:28px;  
}
```

```
.mytext-large-bold{  
font-size:22px;  
font-weight: bold;  
}
```

```
.mytext{  
font-size:18px;  
}
```

```
.mytext-small{  
font-size:15px;  
}
```

```
.mywhite{  
color:white;  
}
```



12. Book.java

```
package com.jlcindia.bookstoreweb;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Book {
    private Integer bookId;
    private String bookName;
    private String author;
    private String publications;
    private String category;

    //Constructors
    //Setters and Getters

}
```

13. BookInfo.java

```
package com.jlcindia.bookstoreweb;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class BookInfo {
    private Integer bookId;
    private String bookName;
    private String author;
    private String publications;
    private String category;
    private double price;
    private int offer;
    private double avgRating;
    private int numberOfSearches;

    //Constructors
    //Setters and Getters

}
```



14. Order.java

```
package com.jlcindia.bookstoreweb;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Order {
    private Integer orderId;
    private String orderDate;
    private String userId;
    private int totalQty;
    private double totalCost;
    private String status;

    //Constructors
    //Setters and Getters

}
```

15. OrderItem.java

```
package com.jlcindia.bookstoreweb;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class OrderItem {
    private Integer orderItemId;
    private Integer orderId;
    private Integer bookId;
    private int qty;
    private double cost;

    //Constructors
    //Setters and Getters

}
```




16. OrderInfo.java

```
package com.jlcindia.bookstoreweb;

import java.util.List;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class OrderInfo {

    private Order order;
    private List<OrderItem> itemsList;

    public OrderInfo() {}
    public OrderInfo(Order order, List<OrderItem> itemsList) {
        super();
        this.order = order;
        this.itemsList = itemsList;
    }

    //Setters and Getters

}
```

17. UserRating.java

```
package com.jlcindia.bookstoreweb;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class UserRating {
    private Integer ratingId;
    private String userId;
    private Integer bookId;
    private double rating;
    private String review;

    //Constructors
    //Setters and Getters

}
```



18. BookStoreController.java

```
package com.jlcindia.bookstoreweb;

import java.util.*;
import javax.servlet.http.*;
import org.slf4j.*;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Controller
public class BookStoreController {

    static Logger log=LoggerFactory.getLogger(BookStoreController.class);

    @Autowired
    BookStoreService bookStoreService;

    private Map<Integer,Book> mycartMap=new LinkedHashMap<>();

    @GetMapping( "/" )
    public String showIndexPage(Model model,HttpSession session) {
        System.out.println("** showIndexPage **");
        List<String> authorsList=bookStoreService.getAuthorsList();
        List<String> catList=bookStoreService.getCategoryList();
        session.setAttribute("MyAuthorList", authorsList);
        session.setAttribute("MyCatList", catList);

        return "redirect:/showAllBooks";
    }

    @GetMapping("/showAllBooks")
    public String showBooksList(HttpServletRequest request, HttpSession session) {
        System.out.println("-----BookStoreController--showBooksList()-----");
        String author=request.getParameter("author");
        String category=request.getParameter("category");
        System.out.println(author);
        System.out.println(category);
    }
}
```



```
Collection<Book> blist=bookStoreService.getMyBooks(author,category);
session.setAttribute("MyBooksList", blist);
session.setAttribute("MyCart", mycartMap);

return "showBooksList";
}

@GetMapping("/showBookInfo")
public String showBookFullInfo(@RequestParam("bookId") String bookId,HttpSession
session,HttpServletRequest request) {
System.out.println("-----BookStoreController--showBookFullInfo()-----");
BookInfo bookInfo=bookStoreService.getBookInfoByBookId(new Integer(bookId));

request.setAttribute("MyBookInfo", bookInfo);
return "showBookInfo";
}

@PostMapping("/addToCart")
public String addBookToCart(@RequestParam("bookId") String bookId,HttpSession
session) {
System.out.println("-----BookStoreController--addBookToCart()-----");
System.out.println(bookId);
Book mybook=bookStoreService.getBookByBookId(new Integer(bookId));
mycartMap.put(new Integer(bookId), mybook);

session.setAttribute("MyCart", mycartMap);
return "showBooksList";
}

@GetMapping("/showMyCart")
public String showBooksInMyCart(HttpServletRequest request,HttpSession session) {
System.out.println("-----BookStoreController--showBooksInMyCart()-----");

Object obj=session.getAttribute("MyCart");
Map<Integer,Book> cartMap=(Map<Integer,Book>)obj;
Collection<Book> cartbookList=(Collection<Book>)cartMap.values();
session.setAttribute("MyCartItem", cartbookList);
if(cartbookList.size()==0) {
request.setAttribute("CartEmptyMSG", "No Books Cart- Please Continue Shopping");
}
return "showCart";
}
```



@PostMapping("/removeFromCart")

```
public String removeBookFromCart(@RequestParam("bookId") String bookId,HttpSession session) {  
    System.out.println("-----BookStoreController--removeBookFromCart()-----");  
    System.out.println(bookId);  
    Object obj=session.getAttribute("MyCart");  
    Map<Integer,Book> cartMap=(Map<Integer,Book>)obj;  
    cartMap.remove(new Integer(bookId));  
  
    return "redirect:/showMyCart";  
}
```

@GetMapping("/continueShopping")

```
public String continueShopping() {  
    System.out.println("-----BookStoreController--continueShopping()-----");  
  
    return "redirect:/showAllBooks";  
}
```

@GetMapping("/placeOrder")

```
public String placeMyOrder( HttpSession session) {  
    System.out.println("-----BookStoreController--placeMyOrder()-----");  
  
    if(mycartMap.size()>0) {  
        bookStoreService.placeOrder(mycartMap);  
        mycartMap.clear();  
    }  
    return "orderSuccess";  
}
```

@GetMapping("/showRatingsForm")

```
public String showRatingsForm(Model model) {  
    System.out.println("-----BookStoreController--showRatingsForm()-----");  
  
    UserRating userRating=new UserRating();  
    userRating.setUserId("U-12");  
  
    model.addAttribute("myUserRating", userRating);  
  
    return "addRating";  
}
```



@PostMapping("/addMyRating")

```
public String addMyRating(@ModelAttribute("") UserRating userRating) {  
    System.out.println("-----BookStoreController--addMyRating()-----");  
    bookStoreService.addUserRating(userRating);  
  
    return "ratingSuccess";  
}
```

@GetMapping("/showMyRatings")

```
public String showMyRatingsList(Model model,HttpSession session) {  
    System.out.println("-----BookStoreController--showMyRatingsList()-----");  
    List<UserRating> userRatingList=bookStoreService.getMyRatings("U-12");  
  
    session.setAttribute("MyUserRatingList", userRatingList);  
  
    return "ratingsList";  
}  
}
```

19. BookStoreService.java

```
package com.jlcindia.bookstoreweb;  
  
import java.util.*;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */  
public interface BookStoreService {  
    public List<String> getAuthorsList();  
    public List<String> getCategoryList();  
  
    public List<Book> getMyBooks(String author, String category);  
    public Book getBookByBookId(Integer bookId);  
    public BookInfo getBookInfoByBookId(Integer bookId);  
  
    public void placeOrder( Map<Integer,Book> mycartMap);  
    public List<Order> getMyOrders(String userId);  
  
    public void addUserRating(UserRating userRating);  
    public List<UserRating> getMyRatings(String userId);  
}
```



20. BookStoreServiceImpl.java

```
package com.jlcindia.bookstoreweb;

import java.text.SimpleDateFormat;
import java.util.*;

import org.slf4j.*;
import org.springframework.stereotype.Service;
import org.springframework.web.client.RestTemplate;

import com.coursecube.bookstoreweb.dto.Book;
import com.coursecube.bookstoreweb.dto.BookInfo;
import com.coursecube.bookstoreweb.dto.Order;
import com.coursecube.bookstoreweb.dto.OrderInfo;
import com.coursecube.bookstoreweb.dto.OrderItem;
import com.coursecube.bookstoreweb.dto.UserRating;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Service
public class BookStoreServiceImpl implements BookStoreService{
    static Logger log=LoggerFactory.getLogger(BookStoreServiceImpl.class);

    Map<Integer,Book> booksMap=new LinkedHashMap<>();

    public List<String> getAuthorsList(){
    List<String> authorsList=new ArrayList<>();
    authorsList.add("All Authors");
    authorsList.add("Srinivas");
    authorsList.add("Vas");
    authorsList.add("Sri");
    return authorsList;
    }

    public List<String> getCategoryList(){
    List<String> catList=new ArrayList<>();
    catList.add("All Categories");
    catList.add("Web");
    catList.add("Spring");
    return catList;
    }
```



```
public List<Book> getMyBooks(String author, String category){
    System.out.println("BookStoreServiceImpl - getBooks()");

    if(author==null || author.length()==0) {
        author="All Authors";
    }
    if(category==null || category.length()==0) {
        category="All Categories";
    }

    //Invoke BookSearchMS
    RestTemplate bookSearchRest=new RestTemplate();
    String endpoint="http://localhost:8000/mybooks/"+author+"/"+category;

    List<Map> list=bookSearchRest.getForObject(endpoint, List.class);
    List<Book> bookList=new ArrayList<>();

    for(Map mymap:list) {
        Book mybook=convertMapToBook(mymap);
        bookList.add(mybook);
        booksMap.put(mybook.getBookId(), mybook);
    }
    return bookList;
}

public BookInfo getBookInfoByBookId(Integer bookId) {
    System.out.println("BookStoreServiceImpl - getBookInfoByBookId()");

    RestTemplate bookSearchRest=new RestTemplate();
    String endpoint="http://localhost:8000/mybook/"+bookId;

    BookInfo bookInfo=bookSearchRest.getForObject(endpoint, BookInfo.class);
    return bookInfo;
}

public Book getBookByBookId(Integer bookId) {
    System.out.println("BookStoreServiceImpl - getBookByBookId()");
    System.out.println(bookId);
    Book mybook=booksMap.get(bookId);

    return mybook;
}
```




```
public void placeOrder( Map<Integer,Book> mycartMap) {

    List<OrderItem> itemList=new ArrayList<>();
    double totalPrice=0.0;
    int totalQuantity=0;

    for(Book mybook:mycartMap.values()) {
        Integer bookId=mybook.getBookId();

        //Invoke BookPrice Controller
        RestTemplate bookPriceRest=new RestTemplate();
        String priceEndpoint="http://localhost:9000/offeredPrice/"+bookId;
        double offerPrice=bookPriceRest.getForObject(priceEndpoint, Double.class);

        OrderItem item=new OrderItem(0, bookId, 1, offerPrice);
        itemList.add(item);

        totalPrice=totalPrice+offerPrice;
        totalQuantity=totalQuantity+1;
    }
    Date today = Calendar.getInstance().getTime();
    SimpleDateFormat formatter = new SimpleDateFormat("dd-MM-yyyy hh:mm");
    String orderDate = formatter.format(today);
    System.out.println(orderDate);

    Order order=new Order(orderDate,"U-111",totalQuantity,totalPrice,"New");

    OrderInfo orderInfo=new OrderInfo();
    orderInfo.setOrder(order);
    orderInfo.setItemsList(itemList);

    //Invoke PlaceOrder MS
    String orderEndpoint="http://localhost:7000/placeOrder";
    RestTemplate orderRest=new RestTemplate();

    orderRest.put(orderEndpoint, orderInfo);
    System.out.println("Order Placed");
}
```




```
public List<Order> getMyOrders(String userId) {  
    // Invoke PlaceOrder MS  
    String orderEndpoint = "http://localhost:7000/myorders/U-123";  
    RestTemplate orderRest = new RestTemplate();  
  
    List<Order> myorders = orderRest.getForObject(orderEndpoint, List.class);  
    //You need to Implement Remaining  
    return myorders;  
}  
  
public void addUserRating(UserRating userRating) {  
    //Invoke UserRating MS  
    String ratingEndpoint="http://localhost:6500/addUserRating";  
    RestTemplate ratingRest=new RestTemplate();  
  
    ratingRest.put(ratingEndpoint, userRating);  
    System.out.println("Rating Added");  
}  
  
public List<UserRating> getMyRatings(String userId){  
  
    List<UserRating> ratingsList=new ArrayList<>();  
  
    String ratingEndpoint="http://localhost:6500/userRatings/"+userId;  
    RestTemplate ratingRest=new RestTemplate();  
  
    List<Map> mymap = ratingRest.getForObject(ratingEndpoint, List.class);  
    for(Map map:mymap) {  
        UserRating urtaing=convertMapToUserRating(map);  
        ratingsList.add(urtaing);  
        System.out.println(map);  
    }  
    return ratingsList;  
}  
  
private UserRating convertMapToUserRating(Map map) {  
    UserRating rating=new UserRating();  
    rating.setRatingId(new Integer(map.get("ratingId").toString()));  
    rating.setUserId(map.get("userId").toString());  
    rating.setBookId(new Integer(map.get("bookId").toString()));  
    rating.setRating(new Double(map.get("rating").toString()));  
    rating.setReview(map.get("review").toString());  
    return rating;  
}
```



```
private Book convertMapToBook(Map map) {  
    Book mybook=new Book();  
    mybook.setBookId(Integer.parseInt(map.get("bookId").toString()));  
    mybook.setBookName((map.get("bookName").toString()));  
    mybook.setAuthor((map.get("author").toString()));  
    mybook.setPublications(map.get("publications").toString());  
    mybook.setCategory(map.get("category").toString());  
    return mybook;  
}  
}
```

21. JLCBookStoreWebConfig.java

```
package com.jlcindia.bookstoreweb;  
  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
import org.springframework.web.servlet.config.annotation.*;  
import org.springframework.context.annotation.*;  
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;  
import org.springframework.web.servlet.view.InternalResourceViewResolver;  
import org.springframework.web.servlet.view.JstlView;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */  
@SpringBootApplication  
public class JLCBookStoreWebConfig implements WebMvcConfigurer {  
  
    @Bean  
    public InternalResourceViewResolver viewResolver() {  
        InternalResourceViewResolver viewResolver = new InternalResourceViewResolver();  
        viewResolver.setViewClass(JstlView.class);  
        viewResolver.setPrefix("/WEB-INF/myjsps/");  
        viewResolver.setSuffix(".jsp");  
        return viewResolver;  
    }  
    @Override  
    public void addResourceHandlers(ResourceHandlerRegistry registry) {  
        registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-INF/resources/webjars/");  
        registry.addResourceHandler("/mycss/**").addResourceLocations("classpath:/META-INF/resources/mycss/");  
    }  
}
```



22.BookStoreWebApplication.java

```
package com.jlcindia.bookstoreweb;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class BookStoreWebApplication {
    static Logger log=LoggerFactory.getLogger(BookStoreWebApplication.class);

    public static void main(String args[]) {
        log.info(" BookStoreWeb - Begin ");
        SpringApplication.run(BookStoreWebApplication.class, args);
        log.info(" BookStoreWeb - End ");
    }
}
```

23. application.properties

```
spring.application.name=MyBookStoreWeb

server.port=5000

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
```

24.pom.xml

```
<project .....">

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookstoreweb</groupId>
<artifactId>BookStoreWeb</artifactId>
<version>1.0</version>
<packaging>jar</packaging>

<name>MyBookStoreWeb</name>
<url>https://www.jlcindia.com</url>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyBookStoreWeb</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

MicroService Name	Rest Endpoints	Used By
MyBookPriceMS	http://localhost:9000/bookPrice/{bookId}	MyBookSearchMS
MyBookPriceMS	http://localhost:9000/offeredPrice/{bookId}	MyBookStoreWeb
MyBookSearchMS	http://localhost:8000/mybooks/{author}/{category}	MyBookStoreWeb
MyBookSearchMS	http://localhost:8000/mybook/{bookId}	MyBookStoreWeb
MyBookSearchMS	http://localhost:8000/updateBookRating	MyUserRatingMS
MyBookSearchMS	http://localhost:8000/updateBookInventory	MyPlaceOrderMS
MyPlaceOrderMS	http://localhost:7000/placeOrder	MyBookStoreWeb
MyPlaceOrderMS	http://localhost:7000/myorders/{userId}	MyBookStoreWeb
MyUserRatingMS	http://localhost:6500/addUserRating	MyBookStoreWeb
MyUserRatingMS	http://localhost:6500/userRatings/{userId}	MyBookStoreWeb